

The minimum number of edges in a pancyclic graph

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September 2026

Abstract

For $n \geq 3$, let $n + h(n)$ be the least number of edges in a graph on n vertices that contains a cycle of every length from 3 to n . Bondy stated in 1971 that $\log_2(n-1) - 1 \leq h(n) \leq \log_2 n + \log_* n + O(1)$, and whether the lower bound can be raised to match is Erdős Problem #1016. We prove that $h(n) = \log_2 n + \log_* n + O(1)$. The key estimate says that in a connected graph of large cycle rank, after removing a union of cycles with few edges and few components, a uniformly random even edge set restricted to the rest is a forest with probability at most $1/2 + o(1)$. This yields a near-halving recurrence for the normalized cycle-length capacity, whose iteration gives the $\log_* n$ term. The proof has been formalized in Lean 4.

MSC 2020: 05C38 (primary); 05C35, 05C50, 60C05 (secondary).

Keywords: pancyclic graphs, cycle space, nonbacktracking matrix, second moment method, Erdős problems.

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1 Introduction

A graph on n vertices is *pancyclic* if it contains a cycle of every length $3, 4, \dots, n$. Let $n + h(n)$ be the least number of edges of a pancyclic simple graph on n vertices, and let $\log_* x$ be the least $j \geq 0$ such that j applications of $\log_2$ take x to a value at most one.

Theorem 1.1. *There is an absolute constant A such that, for every $n \geq 3$,*

$$\log_2 n + \log_* n - A \leq h(n) \leq \lfloor \log_2 n \rfloor + \log_* n + 1.$$

In particular, $h(n) = \log_2 n + \log_ n + O(1)$.*

Bondy [4] stated without proof that $\log_2(n-1) - 1 \leq h(n) \leq \log_2 n + \log_* n + O(1)$. The lower bound is cycle counting: a connected graph with $n+h$ edges has at most $2^{h+1} - 1$ cycles, and a pancyclic graph needs $n-2$ of them. Griffin [7] gave a written proof and computed small values; constructions appear in Sridharan [10], George, Khodkar and Wallis [6], and, for random graphs, Alon and Krivelevich [1]. Whether the lower bound can be raised to $\log_2 n + \log_* n - O(1)$, or even whether $h(n) - \log_2 n \rightarrow \infty$, is Erdős Problem #1016 [3]. Theorem 1.1 answers both questions.

1.1 The main estimate and an outline

Cycle counting ignores that the lengths must fill an interval. That extra rigidity comes from the following estimate. For a graph G , let $\mathcal{Z}(G)$ be its binary cycle space, the even edge sets of G , of dimension $\beta(G) = |E(G)| - |V(G)| + \text{comp}(G)$. Let ξ be uniform on $\mathcal{Z}(G)$, and for an edge set M put

$$P_G(M) = \mathbb{P}(\xi \cap M \text{ is a forest}).$$

Theorem 1.2 (Forest estimate). *There is r_0 such that the following holds for all $r \geq r_0$. Let G be a connected simple graph with $\beta(G) = r$, and let $W \subseteq G$ be a nonempty union of cycles with $m = |E(W)| \leq \sqrt{r}$ edges and $c = \text{comp}(W) \leq (\log_2 r)/100$ components. Then*

$$P_G(E(G) \setminus E(W)) \leq \frac{1}{2} + \frac{20}{\log^{(3)} r}. \quad (1.1)$$

A single parity bit shared by many parts of the graph can make them all acyclic with probability $1/2$, so the threshold is natural. Section 2 shows that zero-cut events of two regions have correlation $2^{\ell-1}$, where ℓ counts the links between them, and that many disjoint small cyclic regions push the forest probability down to about $1/2$.

Section 3 proves Theorem 1.2: we reduce to maximum degree three while keeping W as a marked set with few components, delete short cycles, obtain many moderately long cycles from the nonbacktracking spectrum, and apply the second moment method to the events that such a cycle is a component of ξ . The key point, (3.6), is that two short cycles are linked through at most $c + o(\log r)$ components, so the witness components control all correlations and no expansion is needed.

Section 4 turns Theorem 1.2 into the recurrence

$$\Phi(2^{256R}) \leq \left(\frac{1}{2} + \frac{20}{\log^{(2)} R} \right) \Phi(R) + 2^{2-R}$$

for the normalized cycle-length capacity Φ . Each step costs one exponential level of rank and nearly halves Φ , with summable errors, so $\Phi(r) = O(2^{-\log_* r})$, which is the lower bound. Section 5 gives the construction.

1.2 Formal verification

The proof has been formalized in Lean 4 [5] with Mathlib [9], at <https://github.com/JWKNT/erdos1016>. The version described here is commit `a38913e`: 40,923 lines in 247 active modules, including the entry point, built with Lean v4.19.0 and a pinned Mathlib revision. This commit passed a fresh verification in the repository’s continuous integration, which rebuilds every active project module from source and audits the statements and axioms (GitHub Actions run `36197260250`).

In the `Erdos1016` namespace, `mainTheorem` proves the two-sided asymptotic assertion in Theorem 1.1, with unspecified additive constants; the explicit upper bound is proved by `Problem1016.upper_bound_integer`. Here $h(n)$ is the minimum of $|E(G)| - n$ over pancyclic simple graphs G on the vertex type `Fin n`, and $\log_* n$ is the least k with $n \leq T(k)$, where $T(0) = 1$ and $T(k+1) = 2^{T(k)}$. Theorem 1.2 is `ShortProof.fewComponentForestEstimate`, with the same hypotheses and bound. These theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. Appendix A lists the Lean counterparts of the main steps.

1.3 Conventions

All graphs are finite. Every feasible prescribed-boundary class of edge sets has $2^{\beta(G)}$ members. Regions are induced, $P_G(H) = P_G(E(H))$, and a region is *cyclic* if it contains a cycle. For a region A , δA is its cut, and “ $\delta A = 0$ ” is the event that ξ avoids it. Auxiliary graphs may have loops and parallel edges, with loops counted twice in degrees. All logs are binary except $\ln$; $\log^{(j)} t$ means j successive binary logarithms, and $\log_*$ stops at one.

2 Cut correlations

For a nonnegative random variable Z with $\mu = \mathbb{E}Z > 0$ and a constant $K \geq 0$, Cauchy–Schwarz gives

$$\mathbb{E}Z^2 \leq 2\mu^2 + K\mu \implies \mathbb{P}(Z = 0) \leq 1 - \frac{\mu^2}{\mathbb{E}Z^2} \leq \frac{1}{2} + \frac{K}{4\mu + 2K}. \quad (2.1)$$

Lemma 2.1 (Exceptional cuts). *In a connected graph, let A be a connected region with cut size d . Among connected regions B disjoint from A whose zero-cut events satisfy*

$$\mathbb{P}(\delta A = \delta B = 0) > 2\mathbb{P}(\delta A = 0)\mathbb{P}(\delta B = 0),$$

no vertex-disjoint subfamily has more than d members.

Proof. Contract A and a disjoint subfamily of the B ’s to vertices. The cycle-space projection is onto, so it preserves their joint cut law. For two vertices v, w , count as *links* the vw edges and the components of $G - \{v, w\}$ attached to both. If there are ℓ links, rank subtraction gives

$$\mathbb{P}(\delta v = \delta w = 0) = 2^{\ell-1} \mathbb{P}(\delta v = 0) \mathbb{P}(\delta w = 0). \quad (2.2)$$

Fix v . Append its d incidences as terminal leaves to a spanning forest of $G - v$, and prune nonterminal leaves. Each link to w gives a terminal-containing branch at w . Thus a vertex with at least three links is a branch vertex of this pruned forest, and there are at most d branch vertices. $\square$

Corollary 2.2 (Small cuts). *For $k \geq 1$ vertex-disjoint connected cyclic regions $H_1, \dots, H_k$ with cuts at most d in a connected graph G ,*

$$P_G\left(\bigcup_i E(H_i)\right) \leq \frac{1}{2} + \frac{(d+1)2^d}{2k}. \quad (2.3)$$

Proof. Let E_i say that the i th cut is zero and its inner even set is nonempty. Its probability p_i is at least 2^{-d-1} . Given zero cuts, the inner even sets are independent, so the pairwise correlation ratio is exactly the zero-cut ratio. By Lemma 2.1, each event has at most d exceptional partners in this disjoint family. For the other pairs use $\mathbb{P}(E_i E_j) \leq 2p_i p_j$, and for exceptional partners use $\mathbb{P}(E_i E_j) \leq p_i$. Thus $Z = \sum_i \mathbf{1}_{E_i}$ has

$$\mu \geq k2^{-d-1}, \quad \mathbb{E}Z^2 \leq 2\mu^2 + (d+1)\mu.$$

Each E_i precludes a forest, so (2.1) proves the claim. $\square$

3 Proof of the forest estimate

Let G and W be as in Theorem 1.2, with r large. Put

$$x = \log r, \quad y = \log x, \quad z = \log y, \quad D = \lfloor x/10 \rfloor, \quad L = \lfloor x\sqrt{y} \rfloor,$$

and suppose the probability in (1.1) exceeds $\frac{1}{2} + \varepsilon$, where $\varepsilon = 20/z$. We derive a contradiction. All estimates below are uniform under the two witness bounds.

3.1 Cubicize without disconnecting the witness

Replace a vertex of degree $d > 3$ by a path of $d - 2$ cubic vertices, with two old incidences at each end and one at every internal vertex. If $k \geq 2$ incidences belong to W , put them first along the path and mark the initial segment meeting them. This segment is connected and has at most k vertices. Mark unchanged witness vertices too. In the expanded graph G' , the witness lifts to a subgraph on a marked set P with

$$|P| \leq 2m, \quad \text{comp}(G'[P]) \leq c.$$

Contracting the splitting trees is a cycle-space bijection. An original forest, lifted and augmented by all tree edges, is still a forest; hence restricting to the complement of P can only increase its probability. Prune unmarked leaves and suppress unmarked degree-two vertices. Their even coordinates are respectively zero or equal along a path, so the cycle-space law and the forest event off P are preserved. No marked vertex is removed; suppression may join marked components but cannot split them.

Call the resulting connected multigraph Γ . It has rank r and maximum degree three, every unmarked vertex is cubic, and

$$|P| \leq 2\sqrt{r}, \quad \text{comp}(\Gamma[P]) \leq c, \quad |\Gamma| = 2r + O(\sqrt{r}), \quad P_\Gamma(\Gamma - P) > \frac{1}{2} + \varepsilon. \quad (3.1)$$

The order identity follows from $|\Gamma| = 2r - 2 + \sum_v (3 - \deg_\Gamma v)$, where only marked vertices contribute to the sum.

3.2 Delete short cycles, without connecting them

If $\Gamma - P$ contains $\lceil \sqrt{r} \rceil$ disjoint cycles of length at most D , including loops and digons, their induced regions have cuts at most D , and (2.3) gives $P_\Gamma(\Gamma - P) \leq \frac{1}{2} + O(xr^{-2/5})$, a contradiction. Otherwise choose a maximal such packing and let S be the union of its vertex sets. Then $|S| \leq D\sqrt{r}$, and

$$J = \Gamma - (P \cup S)$$

is simple with girth greater than D , order $v = (2 + o(1))r$, and deficit

$$d_J := \sum_{u \in J} (3 - \deg_J u) \leq 3(|P| + |S|) = O(\sqrt{r} \log r). \quad (3.2)$$

Neither P nor $P \cup S$ is required to be connected.

3.3 Supply cycle weight

For the directed-edge nonbacktracking matrix B , adjacency matrix A_0 and degree matrix Δ of J , put $M(t) = t^2 I - tA_0 + \Delta - I$ (compare the Ihara–Bass identity [2, 8]). Then

$$M(2) = 2(\Delta - A_0) + 3I - \Delta \succeq 0, \quad \mathbf{1}^T M(2 - d_J/v) \mathbf{1} = 0.$$

The least eigenvalue of the real symmetric matrix $M(t)$ is continuous, nonpositive at $2 - d_J/v$, and nonnegative at 2. Hence $M(t)$ is singular for some $t \in [2 - d_J/v, 2]$, with $t > 1$. A null vector u gives the nonzero B -eigenvector $f(i, j) = tu_j - u_i$. Conversely, a nonreal eigenvalue t of B gives $M(t)u = 0$ with $u \neq 0$; multiplication by u^* gives a real quadratic with constant-to-leading ratio at most two, so $|t| \leq \sqrt{2}$. Even powers of real eigenvalues are nonnegative. Therefore

$$2^{-\ell} \operatorname{tr} B^\ell \geq (1 - d_J/(2v))^\ell - 3v 2^{-\ell/2} \quad (\ell \text{ even}).$$

By (3.2), summing over even $6x \leq \ell \leq L$ gives

$$T := \sum_{\ell \leq L} \frac{\operatorname{tr} B^\ell}{2^\ell 2^\ell} \geq (\tfrac{1}{4} + o(1)) \ln \frac{L}{6x} = (\tfrac{1}{8} + o(1)) \ln y.$$

Put $s = \lfloor (D - 1)/2 \rfloor$. Girth makes an s -step nonbacktracking path between fixed end-points unique. Writing $N_t(u, w)$ for the number of ordinary nonbacktracking walks of length t from u to w , we obtain

$$2^{-t} N_t(u, w) \leq \tfrac{3}{2} 2^{-s} \quad (t > s). \quad (3.3)$$

Let $V = \sum_{C \subset J, |C| \leq L} 2^{-|C|}$, over simple cycles. A nonsimple cyclically nonbacktracking walk contains a proper simple cycle; its complementary ordinary closed nonbacktracking walk has length greater than D . Encode by the cycle, an oriented start, the complement and the original root. The factor $1/(2^\ell)$ cancels orientations and root positions; the start vertex and complementary length contribute at most L^2 . Consequently

$$0 \leq T - V \leq \tfrac{3}{2} L^2 2^{-s} V, \quad V \geq (\tfrac{1}{8} + o(1)) \ln y. \quad (3.4)$$

3.4 Normalize cycle indicators

For each cycle $C \subset J$ of length at most L , let I_C say that C is a component of the random even set. Put

$$a_C = 2^{-|C|}, \quad p_C = \mathbb{P}(I_C), \quad X_C = (a_C/p_C) \mathbf{1}_{I_C}, \quad Z = \sum_C X_C.$$

Cubic ambient degree makes I_C exactly the event that all cycle edges are selected. These are feasible binary linear conditions, so $p_C \geq a_C$, $0 \leq X_C \leq 1$ and $\mathbb{E}X_C = a_C$.

For disjoint cycles C, C' , let $\ell(C, C')$ count their direct joining edges and the components of $\Gamma - (V(C) \cup V(C'))$ attached to both. Prescribing the inner cycle words multiplies zero-cut probabilities by separate inner factors. They cancel in the ratio (2.2), giving

$$\mathbb{E}(X_C X_{C'}) = a_C a_{C'} 2^{\ell(C, C')-1}. \quad (3.5)$$

3.5 The deleted cycles count as cyclic components

At most c components of $\Gamma - (V(C) \cup V(C'))$ meet P , by (3.1). Every other component F_i lies in $\Gamma - P$ and has cut d_i ending on the two cycles, with $\sum_i d_i \leq 2L$. Set $t_0 = \lfloor y/2 \rfloor$. At most $2L/t_0$ components have $d_i > t_0$. Among the remaining components, at most $O(t_0 2^{t_0}/\varepsilon)$ are cyclic, by (2.3) and (3.1).

If F_i is a tree, it cannot meet S : any packed cycle meeting F_i is wholly contained in it, since it is disjoint from C, C' . Thus every such tree lies in the high-girth graph J . Cubic ambient degree gives $|F_i| = d_i - 2$. When it meets both cycles and $d_i \leq t_0$, choose a joining path through it of length at most t_0 ; include direct joining edges as paths of length one. These paths have disjoint interiors and distinct endpoints. Partition each cycle into consecutive vertex blocks of size at most $\lfloor D/4 \rfloor$. Two paths joining the same block pair would form a cycle of length at most $2t_0 + D/2 < D$. Hence there are only $O((L/D)^2)$ such paths. Uniformly in the pair,

$$\ell(C, C') \leq c + O\left((L/D)^2 + L/t_0 + t_0 2^{t_0}/\varepsilon\right) = c + o(x). \quad (3.6)$$

3.6 Finish the second moment

Call a disjoint pair exceptional when $\ell(C, C') \geq 3$. By Lemma 2.1, a disjoint packing of exceptional partners of C has at most $|\delta C| \leq L$ members. A maximal packing has union U_C of size at most L^2 , meeting every exceptional partner. Equation (3.3) bounds cycle weight through a fixed vertex by $O(L2^{-s})$. Thus (3.5) and (3.6) give

$$\sum_{C' \text{ exceptional to } C} \mathbb{E}(X_C X_{C'}) \leq a_C 2^{c+o(x)} O(L^3 2^{-s}) = o(1) a_C, \quad (3.7)$$

since $c \leq x/100$ and $s = x/20 + O(1)$. Intersecting distinct cycles are incompatible, nonexceptional disjoint pairs contribute at most $2a_C a_{C'}$, and $\mathbb{E}X_C^2 \leq a_C$. Therefore, with $\mu = \mathbb{E}Z = V$,

$$\mathbb{E}Z^2 \leq 2\mu^2 + (1 + o(1))\mu.$$

Since every C lies in $\Gamma - P$, equations (2.1) and (3.4) yield

$$P_\Gamma(\Gamma - P) \leq \mathbb{P}(Z = 0) \leq \frac{1}{2} + \frac{2 + o(1)}{\ln y} < \frac{1}{2} + \frac{20}{z},$$

contradicting (3.1). This proves Theorem 1.2. $\square$

4 Cycle lengths and the lower bound

Let $\Phi(R)$ be the supremum of $(L-2)2^{-\beta(G)}$ over simple graphs of rank at least R containing every cycle length $3, \dots, L$. Cycle counting gives $\Phi \leq 1$, and vertex identifications make realizers connected.

Theorem 4.1 (Near-halving). *For every sufficiently large integer R ,*

$$\Phi(2^{256R}) \leq \left(\frac{1}{2} + \frac{20}{\log^{(2)} R}\right) \Phi(R) + 2^{2-R}. \quad (4.1)$$

Proof. Consider a connected graph of rank $r \geq 2^{256R}$ containing every cycle length $3, \dots, L$; if $L < 2^{2R} + 1$, then $(L - 2)2^{-r}$ is below the error. Otherwise greedily adjoin a cycle of the least missing length until the union W has rank $t \geq 2R$. At preceding rank $s < 2R$ the added length is at most $2^s + 2$, and at most $\phi 2^s + 3$ when $s \geq R$, where $\phi = \Phi(R)$. Each step increases rank, so at most $2R$ cycles are used and $\text{comp}(W) \leq 2R$. With $m = |E(W)|$,

$$m \leq 2^{2R} + 4R - 1 \leq 2^{3R}, \quad m + 1 \leq \phi 2^{2R} + 2^R + 5R, \quad (m + 1)2^{-t} \leq \phi + 2^{1-R}.$$

Restriction to $M = E \setminus W$ has 2^{r-t} equally likely words. A mixed cycle requires a forest outside word and has at most $m + 1$ lengths per word. Pure outside cycles contribute at most one length per word, and inside cycles at most m lengths. Hence

$$(L - 2)2^{-r} \leq (m + 1)2^{-t} P_G(M) + 2^{-t} + m2^{-r}.$$

Theorem 1.2 applies, since $m \leq \sqrt{r}$ and $\text{comp}(W) \leq 2R \leq (\log r)/100$. As $\log^{(3)} r \geq \log^{(2)} R$, the coefficient of ϕ is at most that in (4.1). Using also $P_G(M) \leq 1$, the remaining terms are at most

$$2^{1-R} + 2^{-2R} + 2^{3R-r} \leq 2^{2-R}$$

for large R , proving (4.1). $\square$

Corollary 4.2 (Lower bound). *Every pancyclic graph on n vertices has at least $n + \log n + \log_* n - O(1)$ edges.*

Proof. Choose an integer R_0 large enough for Theorem 4.1, and iterate $R_{j+1} = 2^{256R_j}$. For $j \geq 2$, $\log^{(2)} R_j = 8 + 256R_{j-2}$, so

$$\sum_j \frac{1}{\log^{(2)} R_j} < \infty \quad \text{and} \quad \sum_j 2^{j-R_j} < \infty,$$

and (4.1) yields $\Phi(R_j) = O(2^{-j})$.

Also $\log_* R_j = j + O(1)$: compare with $U_{j+1} = 2^{U_j}$ and induct on $R_j \leq \sqrt{U_j}$, starting with a large fixed U_0 ; the reverse bound follows from $R_{j+1} \geq 2^{R_j}$. Monotonicity then gives

$$\Phi(r) = O(2^{-\log_* r}).$$

A pancyclic n -vertex graph with $n + h$ edges has rank $r = h + 1$. Hence $n - 2 = O(2^{r - \log_* r})$ and $n \leq 2^{r+1}$, and thus $\log_* n \leq \log_* r + 2$. These inequalities imply $h \geq \log n + \log_* n - O(1)$. $\square$

5 The matching construction

Proposition 5.1. *For every $n \geq 3$ there is a pancyclic graph on n vertices with at most $n + \lfloor \log n \rfloor + \log_* n + 1$ edges.*

Proof. For $3 \leq n \leq 6$, an n -cycle $v_0 \cdots v_{n-1}$ with the chord $v_0 v_2$ (for $n \geq 4$) and also $v_0 v_3$ (for $n = 6$) suffices.

For $n \geq 7$, choose the largest k with $2^k + k + 1 \leq n$. On an n -cycle, an arc with successive segment lengths $2^i + 1$, $0 \leq i < k$, has endpoints $v_0, \dots, v_k$. Add the k segment shortcuts and $v_0 v_k$. Binary subset sums give the cycle lengths

$$[k + 1, 2^k + k] \quad \text{and} \quad [n - 2^k + 1, n].$$

Maximality of k gives $n \leq 2^{k+1} + k + 1$, so at most one intermediate length is missing; one additional chord supplies it if needed.

Starting at $t = k$, while $t > 2$ add $v_0 v_j$ for $j = \lceil \log t \rceil < t$. Its shortcuts give $[j + 1, 2^j + j]$; since $2^j \geq t$, this extends coverage down to $j + 1$. Replace t by j . This takes $\log_* k - 1$ further chords, so altogether at most

$$k + \log_* k + 1 \leq \lfloor \log n \rfloor + \log_* n + 1$$

edges are added. □

Corollary 4.2 and Proposition 5.1 prove Theorem 1.1.

A Lean correspondence

Module paths are relative to `Erdos1016/` at commit `a38913e`; declaration names below omit namespace prefixes. The formal proof follows the same argument, with explicit constants in place of some O -terms and minor differences in auxiliary rounding and intermediate estimates. The stated forest bound and final upper bound agree exactly.

Paper	Module	Declaration
Thm 1.1	<code>Main.lean</code>	<code>mainTheorem</code>
Lemma 2.1	<code>CycleSpace/Graphical/</code> <code>MultigraphCutCorrelation.lean</code> , <code>CycleSpace/Graphical/</code> <code>DisjointRegionCuts.lean</code>	<code>multigraph_zeroCut_pair</code> , <code>exceptional_partner_card_le</code>
Cor. 2.2	<code>Probability/Moments/</code> <code>SmallCutRegionBound.lean</code>	<code>smallCut_forest_bound</code>
§3.1	<code>Graph/Cubicization/</code> <code>MarkedReduction.lean</code>	<code>exists_marked_subcubic_reduction</code>
§3.2	<code>Cycles/Selection/</code> <code>ShortRegionPacking.lean</code> , <code>Cycles/</code> <code>Selection/RetainedHighGirth.lean</code>	<code>retained_girth</code>
§3.3	<code>Nonbacktracking/Spectrum/</code> <code>DeficitTrace.lean</code> , <code>Cycles/Counting/</code> <code>LowDegreeTraceCycles.lean</code>	<code>normalized_even_trace_lower</code> , <code>cycleWordMass_ge_log_ratio</code>
§3.4	<code>Probability/Moments/</code> <code>SeedPairCorrelation.lean</code>	<code>normalizedSelectedSeed_pair</code>
§3.5	<code>Cycles/Geometry/</code> <code>ActualCycleLinkBound.lean</code>	<code>actual_cycle_links_le</code>

Paper	Module	Declaration
§3.6	Probability/Moments/ SeedFamilyAvoidance.lean, Probability/Moments/ InducedCycleAvoidance.lean	second_moment_le, forest_le_half_add_twenty
Thm 1.2	Probability/Avoidance/ FewComponentForest.lean	fewComponentForestEstimate
Thm 4.1	Extremal/Witness/CapacityBudget. lean, Extremal/Recurrence/ WitnessReduction.lean	exists_budgeted_witness, recurrence_of_few_component_ forest_estimate
Cor. 4.2	Extremal/Recurrence/ LinearExponentialDecay.lean	recurrence_logStar_bound
Prop. 5.1	Extremal/UpperBound.lean	upper_bound_integer

References

- [1] Y. Alon and M. Krivelevich, *Sparse pancyclic subgraphs of random graphs*, SIAM J. Discrete Math. **39** (2025), 562–574. [arXiv:2308.01564](#).
- [2] H. Bass, *The Ihara–Selberg zeta function of a tree lattice*, Internat. J. Math. **3** (1992), 717–797. [doi:10.1142/S0129167X92000357](#).
- [3] T. F. Bloom, *Erdős Problems*, Problem #1016, <https://www.erdosproblems.com/1016>, accessed September 2026.
- [4] J. A. Bondy, *Pancyclic graphs I*, J. Combin. Theory Ser. B **11** (1971), 80–84. [doi:10.1016/0095-8956\(71\)90016-5](#).
- [5] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635. [doi:10.1007/978-3-030-79876-5_37](#).
- [6] J. C. George, A. Khodkar and W. D. Wallis, *Pancyclic and Bipancyclic Graphs*, SpringerBriefs in Mathematics, Springer, 2016. [doi:10.1007/978-3-319-31951-3](#).
- [7] S. Griffin, *Minimal pancyclicity*, [arXiv:1312.0274](#) (2013).
- [8] M. Kotani and T. Sunada, *Zeta functions of finite graphs*, J. Math. Sci. Univ. Tokyo **7** (2000), 7–25.
- [9] The mathlib Community, *The Lean mathematical library*, in: CPP 2020, ACM, 2020, 367–381. [doi:10.1145/3372885.3373824](#).
- [10] M. R. Sridharan, *On an extremal problem concerning pancyclic graphs*, J. Math. Phys. Sci. **12** (1978), 297–306.